Spur Gears Component Generator (Version: 2025.2 (Build 292293000, 293))

5/28/2025

Project Info (iProperties)

Guide

Design Output - Total Unit Correction
Unit Correction Distribution - Custom
Type of Load Calculation - Torque calculation for the specified power and speed
Type of Strength Calculation - Check Calculation
Method of Strength Calculation - ANSI/AGMA 2001-D04:2005

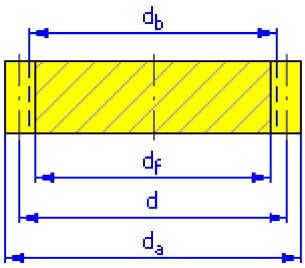
Common Parameters

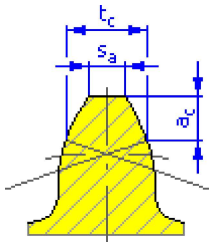
Gear Ratio	i	0.2222 ul
Desired Gear Ratio	i_{in}	0.2222 ul
Module	m	10.000 mm
Helix Angle	β	0.0000 deg
Pressure Angle	α	20.0000 deg
Center Distance	a_w	660.000 mm
Product Center Distance	a	660.000 mm
Total Unit Correction	Σx	0.0000 ul
Circular Pitch	p	31.416 mm
Base Circular Pitch	p_{tb}	29.521 mm
Operating Pressure Angle	α_w	20.0000 deg
Contact Ratio	ε	1.7313 ul
Limit Deviation of Axis Parallelity	f_x	0.0200 mm
Limit Deviation of Axis Parallelity	f_y	0.0100 mm

Gears

		Gear 1	Gear 2
Type of model		Component	Component
Number of Teeth	z	108 ul	24 ul
Unit Correction	x	0.0000 ul	0.0000 ul
Pitch Diameter	d	1080.000 mm	240.000 mm
Outside Diameter	d_a	1100.000 mm	260.000 mm
Root Diameter	d_f	1055.000 mm	215.000 mm
Base Circle Diameter	d_b	1014.868 mm	225.526 mm
Work Pitch Diameter	d_w	1080.000 mm	240.000 mm
Facewidth	b	120.000 mm	120.000 mm
Facewidth Ratio	b_r	0.1111 ul	0.5000 ul
Addendum	a*	1.0000 ul	1.0000 ul
Clearance	c*	0.2500 ul	0.2500 ul
Root Fillet	r_f*	0.3500 ul	0.3500 ul
Tooth Thickness	s	15.708 mm	15.708 mm

Tangential Tooth Thickness	s_t	15.708 mm	15.708 mm
Chordal Thickness	t_c	13.870 mm	13.870 mm
Chordal Addendum	a_c	7.476 mm	7.476 mm
Chordal Dimension	W	384.142 mm	77.165 mm
Chordal Dimension Teeth	z_w	13.000 ul	3.000 ul
Dimension Over (Between) Wires	M	1107.185 mm	266.462 mm
Wire Diameter	d_M	18.000 mm	18.000 mm
Limit Deviation of Helix Angle	F_β	0.0200 mm	0.0170 mm
Limit Circumferential Run-out	F_r	0.0610 mm	0.0300 mm
Limit Deviation of Axial Pitch	f_{pt}	0.0150 mm	0.0110 mm
Limit Deviation of Basic Pitch	f_{pb}	0.0140 mm	0.0100 mm
Virtual Number of Teeth	z_v	108.000 ul	24.000 ul
Virtual Pitch Diameter	d_n	1080.000 mm	240.000 mm
Virtual Outside Diameter	d_{an}	1100.000 mm	260.000 mm
Virtual Base Circle Diameter	d_{bn}	1014.868 mm	225.526 mm
Unit Correction without Tapering	x_z	-2.2149 ul	0.3325 ul
Unit Correction without Undercut	x_p	-5.2971 ul	-0.3840 ul
Unit Correction Allowed Undercut	x_d	-5.4670 ul	-0.5540 ul
Addendum Truncation	k	0.0000 ul	0.0000 ul
Unit Outside Tooth Thickness	s_a	0.8097 ul	0.7156 ul
Tip Pressure Angle	α_a	22.6897 deg	29.8411 deg





Loads

		Gear 1	Gear 2
Power	P	55.000 kW	55.000 kW
Speed	n	95.00 rpm	427.50 rpm
Torque	T	5528.540 N m	1228.564 N m
Efficiency	η	1.000 ul	
Radial Force	F_r	3726.341 N	
Tangential Force	F_t	10238.037 N	
Axial Force	F_a	0.000 N	
Normal Force	F_n	10895.092 N	
Circumferential Speed	v	5.372 mps	
Resonance Speed	n_{E1}	939.338 rpm	

Material

	Gear 1	Gear 2
	AISI 4140	AISI 4140

Ultimate Tensile Strength	S_u	1130 MPa	1130 MPa
Yield Strength	S_y	885 MPa	885 MPa
Modulus of Elasticity	E	195000 MPa	195000 MPa
Poisson's Ratio	μ	0.290 ul	0.290 ul
Allowable Bending Stress	s_{at}	380.2 MPa	413.9 MPa
Allowable Contact Stress	s_{ac}	1539.1 MPa	1539.1 MPa
Hardness in Tooth Core	JHV	210 ul	210 ul
Type of Treatment	type	4 ul	4 ul

Strength Calculation

Factors of Additional Load

Overload Factor	K_o	1.000 ul	
Dynamic Factor	K_v	1.079 ul	
Size Factor	K_s	1.000 ul	1.000 ul
Reliability Factor	K_R	0.850 ul	
Temperature Factor	k_t	1.000 ul	
Load Distribution Factor	K_m	1.278 ul	1.278 ul
Lead Correction Factor	C_{mc}	1.000 ul	1.000 ul
Mesh Alignment Correction Factor	C_e	1.000 ul	
Pinion Proportion Modifier	C_{pm}	1.100 ul	
Mesh Alignment Factor	C_{ma}	Commercial Enclosed Gear Units (0.1992)	

Factors for Contact

Surface Condition Factor	C_f	1.000 ul	1.000 ul
Stress Cycle Factor	Z_N	0.907 ul	0.834 ul
Hardness Ratio Factor	C_H	1.000 ul	1.000 ul
Elastic Factor	C_p	2216.888 ul	
Geometry Factor	I	0.116 ul	

Factors for Bending

Reverse Loading Factor	Y_a	1.000 ul	1.000 ul
Rim Thickness Factor	K_B	1.000 ul	1.000 ul
Stress Cycle Factor	Y_N	0.945 ul	0.901 ul
Geometry Factor	J	0.584 ul	0.569 ul

Results

Factor of Safety from Pitting	k_f	4.346 ul	3.995 ul
Factor of Safety from Tooth Breakage	k_n	20.993 ul	21.223 ul
Check Calculation		**Positive**	

Summary of Messages

5:55:09 AM Design: Numbers of teeth are commensurable - shots of the same teeth are taken relatively regularly
5:55:09 AM Design: Gear 2: The Unit Correction (x) is less than the Unit Correction without Tapering (x_z)

5:55:09 AM Calculation: Calculation indicates design compliance!